Algebra and Discrete Mathematics (ADM)

Tutorial 5 Vector spaces and linear independence

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Vector space of polynomials

- Let $\mathbb{R}[x]$ denote the set of polynomials with coefficients from $\mathbb{R}$, i.e.

$$\mathbb{R}[x] = \left\{ \sum_{i=0}^{n} a_i x^i \;\middle|\; a_i \in \mathbb{R}, n \geq 0 \right\}$$

- An element of $\mathbb{R}[x]$ is of the form
  $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{R}[x]$
- If $a_n \neq 0$, we define *degree of* $f(x)$, denoted $\deg(f(x))$, to be n.
- By convention, we define $\deg(0) = -\infty$.

Vector space of polynomials

$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$,
$g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$ in $\mathbb{R}[x]$
Without loss of generality, let us assume $n \geq m$, write

$$g(x) = b_n x^n + b_{n-1}x^{n-1} + \cdots + b_0,$$

where $b_i = 0$ for $i > m$. Then

$$f(x) + g(x) := c_n x^n + c_{n-1}x^{n-1} + \cdots + c_0, \text{ where } c_i = a_i + b_i.$$

For any $\alpha \in \mathbb{R}$, define scalar multiplication by

$$\alpha f(x) := \alpha a_n x^n + \alpha a_{n-1}x^{n-1} + \cdots + \alpha a_0.$$

Thus $(\mathbb{R}[x], +, \cdot)$ is a vector space over $\mathbb{R}$.

Vector space of polynomials

$$\mathbb{R}[x] = \left\{ \sum_{i=0}^{n} a_i x^i \;\middle|\; a_i \in \mathbb{R}, n \geq 0 \right\}$$

- $(\mathbb{R}[x], +, \cdot)$ is a vector space
- Additive identity/zero vector: 0
- Additive inverse of $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ is

$$-a_n x^n - a_{n-1} x^{n-1} - \cdots - a_0$$

- Other axioms also hold, following from the properties of real numbers

Linear independent vectors

Given the vectors:

$$\boldsymbol{u} = \begin{pmatrix} -1, & 2, & -3 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} 1, & 0, & 4 \end{pmatrix}, \quad \boldsymbol{w} = \begin{pmatrix} 2, & -2, & 1 \end{pmatrix}$$

Are these vectors linearly dependent or independent?
We compute the determinant:

$$\begin{vmatrix} -1 & 1 & 2 \\ 2 & 0 & -2 \\ -3 & 4 & 1 \end{vmatrix}$$

Apply Sarrus' rule

$$0 + 16 + 6 - (0 + 8 + 2) = 12 \neq 0$$

Since the determinant is nonzero, the vectors are **linearly independent**.

Linearly independent vectors

Given the vectors:

$$\boldsymbol{u} = \begin{pmatrix} 2, & 3 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} 5, & -1 \end{pmatrix}, \quad \boldsymbol{w} = \begin{pmatrix} 1, & 4 \end{pmatrix}$$

Are these vectors linearly dependent?
Consider

$$\alpha_1 \begin{pmatrix} 2, & 3 \end{pmatrix} + \alpha_2 \begin{pmatrix} 5, & -1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1, & 4 \end{pmatrix} = \begin{pmatrix} 0, & 0 \end{pmatrix},$$

which corresponds to the following homogeneous linear system

$$\begin{aligned} 2\alpha_1 + 5\alpha_2 + \alpha_3 &= 0 \\ 3\alpha_1 - \alpha_2 + 4\alpha_3 &= 0 \end{aligned}$$

Recall

Corollary

A homogeneous linear system with more unknowns than equations has infinitely many solutions.

The system has nontrivial solutions, the vectors are **linearly dependent**.

Linearly independent vectors

Are the following vectors linearly independent?

$$\boldsymbol{v}_1 = \begin{pmatrix} 1, & 3, & 5, & 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} -1, & 2, & 1, & 1 \end{pmatrix},$$

$$\boldsymbol{v}_3 = \begin{pmatrix} 2, & 2, & 0, & -1 \end{pmatrix}, \quad \boldsymbol{v}_4 = \begin{pmatrix} 1, & 0, & 1, & 0 \end{pmatrix}$$

Cofactor expansion along the fourth row, and Sarru's rule

$$\begin{aligned} \begin{vmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 5 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{vmatrix} &= (-1)^{4+2} \begin{vmatrix} 1 & 2 & 1 \\ 3 & 2 & 0 \\ 5 & 0 & 1 \end{vmatrix} - (-1)^{4+3} \begin{vmatrix} 1 & -1 & 1 \\ 3 & 2 & 0 \\ 5 & 1 & 1 \end{vmatrix} \\ &= (2 - 10 - 6) - (2 + 3 - 10 + 3) \\ &= -14 - 2 = -16 \neq 0 \end{aligned}$$

Therefore, the vectors are linearly independent.

Linear combination

$$\boldsymbol{u} = \begin{pmatrix} 1, & 2, & 3 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} 3, & 2, & 1 \end{pmatrix}, \quad \boldsymbol{w} = \begin{pmatrix} 6, & 8, & 10 \end{pmatrix}$$

Is $\boldsymbol{w}$ a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$

$$\begin{vmatrix} 1 & 3 & 6 \\ 2 & 2 & 8 \\ 3 & 1 & 10 \end{vmatrix} = 0 \Longrightarrow \text{the three vectors are linearly dependent}$$

$$\alpha_1 \boldsymbol{u} + \alpha_2 \boldsymbol{v} = \boldsymbol{w}$$

$$\begin{aligned} \alpha_1 + 3\alpha_2 &= 6 \\ 2\alpha_1 + 2\alpha_2 &= 8 \\ 3\alpha_1 + \alpha_2 &= 10 \end{aligned}$$

Solve by, e.g. Cramer's rule using the first two equations

$$\alpha_1 = 3, \quad \alpha_2 = 1$$

These values also satisfy the third equation, so $\boldsymbol{w} = 3\boldsymbol{u} + \boldsymbol{v}$.

Linearly independent matrices

Are the three matrices linearly independent in $\mathcal{M}_{2\times 2}$?

$$M_1 = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$$

$$\alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 M_3 = O$$

gives

$$\begin{pmatrix} \alpha_1 & 0 \\ \alpha_1 & 2\alpha_1 \end{pmatrix} + \begin{pmatrix} \alpha_2 & 2\alpha_2 \\ 2\alpha_2 & \alpha_2 \end{pmatrix} + \begin{pmatrix} 0 & \alpha_3 \\ 2\alpha_3 & \alpha_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned} \alpha_1 + \alpha_2 &= 0 \\ 2\alpha_2 + \alpha_3 &= 0 \\ \alpha_1 + 2\alpha_2 + 2\alpha_3 &= 0 \\ 2\alpha_1 + \alpha_2 + \alpha_3 &= 0 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 2 & 2 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right)$$

Linearly independent matrices

Are the three matrices linearly independent in $\mathcal{M}_{2\times 2}$?

$$M_1 = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 \end{array}\right) \xrightarrow[R_3 \to R_3 - R_1]{R_4 \to R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \xrightarrow[R_3 \to R_3 - 2R_2]{R_4 \to R_4 + R_2}$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \xrightarrow{R_4 \to R_4 + R_3} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Longrightarrow \text{only trivial solution}$$

$\Longrightarrow$ the matrices are linearly independent